Q-KNEE QUANTUM DIAGNOSTICS CENTER

Quantum-Assisted MRI Analysis Report (Page 1 of 2)

Quantum-Assisted Musculoskeletal MRI Screening — research prototype, not a certified medical device.

Generated: 2026-08-31 09:13:34 UTC

Study ID (De-identified): DEMO-001

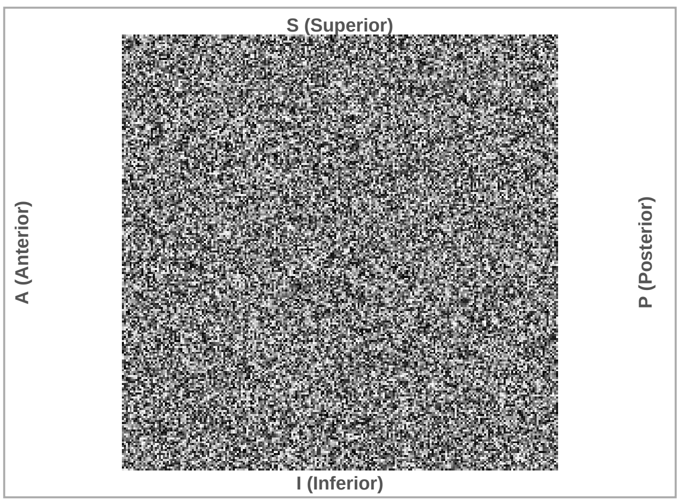
Clinical Impression

Overall Risk Score	**72.0%**	HIGH
Predicted Condition	**ACL Tear**	
Confidence Interval (heuristic ±8pp)	64.0%–80.0%	

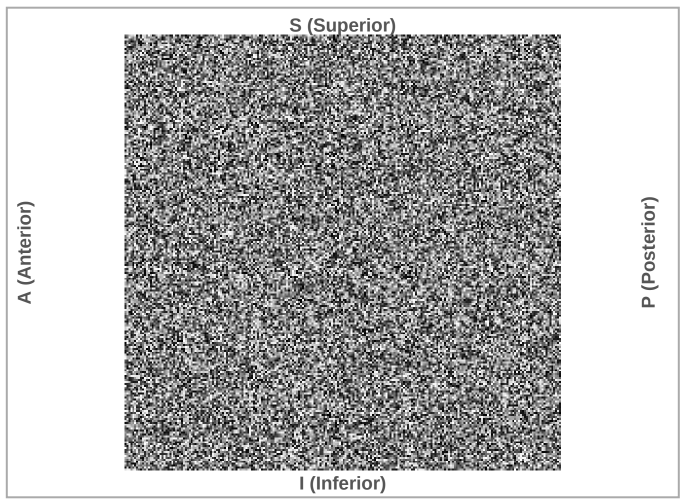
Patient & Study Information

Patient ID	DEMO-001	**Scan Date**	2026-08-31
Patient Name	Test Patient	**Modality**	MRI Knee
Date of Birth	1990-01-01	**Referring Physician**	Dr. A. Example
Clinical Indication	Chronic knee pain, rule out ACL/meniscal tear		

Visual Evidence — MRI Slice & Risk-Targeted Grad-CAM Overlay



Original MRI Slice



Grad-CAM Risk-Region Overlay

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Quantum Feature Attribution — 4-Qubit VQC Measurement

Qubit	Pauli-Z Expectation <Z>	Feature Impact Weight	Weighted Contribution
Q0	+0.4200	+0.8500	+0.3570
Q1	-0.1800	-0.4000	+0.0720
Q2	+0.6300	+1.1000	+0.6930
Q3	-0.0700	+0.2500	-0.0175

Each qubit's Pauli-Z expectation <Z> (measurement basis, range [-1, 1]) and, when available, its trained readout weight and weighted contribution to the pre-sigmoid risk logit.

Diagnostic Breakdown

Region	Tear Risk	95% CI	Risk Tier	Classification
ACL	72.0%	64.0%–80.0%	**HIGH**	TEAR LIKELY
MCL	44.0%	36.0%–52.0%	**MODERATE**	NO TEAR INDICATED
Meniscus	31.0%	23.0%–39.0%	**LOW**	NO TEAR INDICATED

Quantum Circuit Inference Latency

Feature Extraction	Dim. Reduction	Quantum Circuit Inference	Total (End-to-End)
18.40 ms	1.20 ms	42.70 ms	62.30 ms

Inference backend: live